

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Branch: Mechanical Engineering/Automobile Engineering/Production Engineering

Course: TY BTech

Semester: VI

Subject Code & Name: BTMOE605D Wind Energy

Max Marks: 60

Date: 30/12/2024

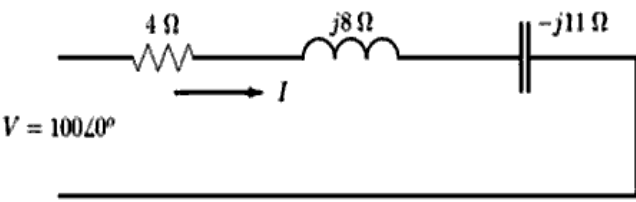
Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The wind speed is measured using an instrument called a. hydrometer b. Manometer c. anemometer d. wind vane	CO1	1
2	Which type of windmill has better performance ? a. Vertical type wind mills b. Darrieus type machines c. Magnus effect rotor d. Horizontal type windmills	CO1	1
3	How much is the energy available in the winds over the earth surface is estimated to be ? a. 2.9 X 120 MW b. 1.6 X 107 MW c. 1 MW d. 5MW	CO3	1
4	How much power does the small scale wind machine generate? a. 18 KW b. 2 KW c. 12 KW d. 30 KW	CO3	1
5	What are the components of a generator? a. The major components of a generator include the rotor, shaft, and load. b. The major components of a generator include the field winding, armature winding, and slip rings. c. The major components of a generator include the stator, rotor, and the shaft. d. The major components of a generator include the turbine, gearbox, and generator.	CO2	1
6	What is the basic mode of operation referred to for a self-excited dc generator?	CO2	1

	a. asynchronous mode	b. self-excited mode	synchronous mode	d. permanent magnet mode		
7	What quantity produces magnetic flux in a permanent magnet generator?				CO2	1
	a. permanent magnets	b. armature reaction	c. residual magnetism	d. field current		
8	What effect causes the generated emf E to be greater than zero for a dc shunt generator even when the field current is negligible?				CO2	1
	a. saturation of the magnetic circuit	b. armature reaction	c. residual magnetism of the poles	d. self-excitation		
9	Why is it desirable to maintain the three line currents of a three-phase generator at about the same value ?				CO2	1
	a. maximize output power	b. improve power factor	c. to minimize torque fluctuations	d. balance the voltages		
10	What is one limitation to decreasing the field resistance R_f of a dc shunt generator ?				CO2	1
	a. saturation limits	b. armature reaction limits	c. R_f can not be reduced below the internal coil resistance	d. rectifier voltage drop limits		
11	What are the two modes through which wind generated electricity has value according to the text ?				CO3	1
	a. From capacity credits and plant factor	b. From fuel costs and capacity factor	c. From capacity displacement of conventional generation and from saving fuel	d. From capacity credits and fuel savings		
12	What are the two common options a utility may consider when planning new generation that wind generation could potentially displace?				CO3	1
	a. New coal generation or to save oil at existing oil fired units					
	b. New hydroelectric generation or save geothermal at existing geothermal units					
	c. New solar generation or save biomass at existing biomass units					
	d. New nuclear generation or save natural gas at existing natural gas unit.					
Q. 2	Solve the following.					12
A)	Explain the horizontal axis wind turbine with suitable schematic diagram.				CO4	6
B)	Explain the Weibull statistics for wind measurement.				CO4	
Q.3	Solve the following.					12
A)	Derive and explain the power output from a practical wind turbine.				CO5	6
B)	Explain the terms for wind turbine, i. tip speed ratio, ii. capacity factor				CO4	6

Q. 4	Solve Any Two of the following.		12
A)	Explain the capacity credit for the wind energy power plant.	CO4	6
B)	Explain the working of three phase generator with schematic diagram.	CO5	6
C)	A series RLC circuit, shown in Fig., has $R=4\Omega$, $X_L=8\Omega$, and $X_C=11\Omega$. Find the current, complex power, real power, and reactive power delivered to the circuit for an applied voltage of 100 V. What is the power factor? 	CO6	6
Q.5	Solve Any Two of the following.		12
A)	Explain the operation of single phase induction generator used in wind energy power generation.	CO4	6
B)	List out the various types of asynchronous loads and explain in detail any one of it.	CO5	6
C)	A piston pump has an area A of 0.01 m^2 and a stroke of 0.2 m . The manufacturers data sheet lists a recommended maximum operating speed of 50 cycles per minute. The slip is estimated as 0.05. What is the pump volume capacity ?	CO6	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the in detail about the various parameters considered for the estimation of capital cost of wind energy plant.	CO4	6
B)	Define and explain the following terms with respect to the cost of wind turbine. i. Total cost, ii. Cost per unit area, iii. Cost per kW of rating, iv. Unit cost of electricity	CO5	6
C)	An Enertech 4000 reaches a maximum power of 4.2 kW in an 11-m/s wind speed. The propeller diameter is 6 m. The installed cost at your location is Rs. 10,000 in 1981. You estimate the Weibull parameters at your site to be $c=8\text{ m/s}$ and $k=2.2$. The interest rate is 0.11 and the period of the loan or the desired payback period is 15 years. Find C_a , C_{kW} , and C_U . Do not include factors such as tax credits, inflation, or operation and maintenance costs.	CO6	6
*** End ***			